



PayNet Achieves Near Real-Time E-Payment Settlement Views with Sourced Group and AWS

PayNet delivers near real-time views of digital banking settlement positions by decoupling settlement reporting services and migrating to a modern data platform on AWS, with support from AWS Partner Sourced Group.

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Near real-time views

of digital banking settlement position:



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by analyzing volumes of data

Hourly



Overview

[Payments Network Malaysia Sdn Bhd](#) (PayNet) provides the national payments network for Malaysia. With e-payment volumes accelerating, the organization built a cloud-based data platform on Amazon Web Services (AWS) to improve settlement reporting and unlock value from accumulated data.

PayNet implemented [Amazon Managed Streaming for Apache Kafka](#) and [Amazon Elastic Kubernetes Service](#) to balance the data platform's cost with processing performance. As a result, PayNet supports near real-time views of settlement positions and ingests millions of e-payment transactions daily.



Opportunity | Unlocking Value from Growing E-Payment Transactions

Payments Network Malaysia (PayNet) is the national payments network and shared infrastructure for Malaysia's financial markets. The organization is jointly owned by 11 of Malaysia's major banks; the country's central bank, Bank Negara Malaysia (BNM), is the largest shareholder.

PayNet is dedicated to advancing Malaysia's digital economy by promoting the use of electronic payments (e-payments) to support the nation's pursuit of efficient, low-cost, and secure financial services. Among PayNet's offerings is DuitNow, facilitating instant transfers to recipients' DuitNow IDs, such as their mobile or national identification numbers. Other services include MyDebit, Malaysia's national debit card scheme for purchases at various merchants, FPX, an online payment gateway facilitating real-time debitir



transactions, and JomPAY, simplifying bill payments through internet and mobile banking. PayNet also provides additional services like Interbank GIRO (IBG), Direct Debit, and a Shared ATM Network (SAN).

In line with its commitment, PayNet aimed to enhance its capabilities as a data-driven organization to expedite the transition to e-payments, promoting financial inclusion by bridging the gap between the banked and unbanked populations. This initiative fosters economic growth and innovation while decreasing dependence on cash. Therefore, increasing the frequency of digital transaction settlements for participating banks is vital, as is providing near real-time views of settlement positions for banks to mitigate risks, enhance liquidity management, and strengthen the country's overall risk posture.

To do this, PayNet sought to migrate resource-intensive processing services to a high-performance cloud-based data analytics platform. With a modern data architecture, the organization could bolster its analytics capabilities to support multiple pipelines through massive parallel processing, data products, and consumption models. The company decided to design, build, and deploy the data analytics platform on Amazon Web Services (AWS).

Solution | Decoupling Reporting and Analytics Services with a Modern Data Platform

PayNet turned to [AWS Partner Sourced Group](#) (Sourced), a global cloud consulting firm which PayNet has worked with for over two years. Engineering teams from Sourced and PayNet collaborated to develop the platform's data lake and the multiple extract, transform, and load (ETL) pipelines that serve as reporting and analytics information sources.

After subsequent deep dives into the platform's data requirements, PayNet and Sourced engineers created the pipelines to make the ETL highly efficient and streamlined. The teams followed AWS best practices, ensuring roles and access permissions were segregated between data producers and consumers. PayNet and Sourced also implemented the data landing zone and the raw, transformed, and consumption layers of the data architecture. Kenny Tse, head of Data Lake at PayNet says, "We knew Sourced is a trusted partner that PayNet could collaborate with to deliver our data analytics goals through its AWS expertise and experience."

To discover, prepare, and integrate data from multiple pipelines, PayNet used Amazon Managed Streaming for Apache Kafka (Amazon MSK) for real-time streaming events with Amazon Elastic Kubernetes Service (Amazon EKS)



to help sync data on the platform. Dickson Tung, lead for Data Platform Engineering at PayNet, says, "The decision to opt for Amazon MSK primarily hinges on cost benefits associated with real-time streaming of transaction data. Meanwhile, Amazon EKS provides high availability and platform resiliency, ensuring optimal operation of self-healing applications."

After implementing the platform's core elements, PayNet deployed [Amazon QuickSight](#), a unified business intelligence service, to provide reports and business insights via dashboards to internal data consumers across the organization.

The platform was launched in May 2023 and progressively integrated with various PayNet products and services, including DuitNow, MyDebit, and FPX. Currently, it processes over 20 million e-payment transactions daily, with this number continuing to increase. Loi, lead for Data Engineering & Analytics at PayNet, adds, "As our AWS data lake expands rapidly, implementing the right data platform architecture has effectively managed costs, emphasizing the value of adopting a cost-effective solution on AWS."

Outcome | Reducing Settlement Views to Near Real-Time on AWS

By migrating reporting services to a modern data platform on AWS, PayNet can provide views of e-payment settlement positions much faster. This, in turn, is helping accelerate the growth of e-payment transactions, expected to [average 400 per person per year by 2026](#). Tse adds, "By building our data platform with AWS and Sourced, we've achieved near real-time views on e-payment settlements, significantly reducing risk, uncovering new opportunities, and enhancing operational support. He continues, "We now have the potential to increase settlement frequency across all our offerings, delivering a better service to banks and driving digital banking services across Malaysia."

Today, PayNet can also sift through huge amounts of data to bolster fraud detection and prevention. "Now that we can ingest larger volumes of transaction data in real-time with our AWS platform, designed and deployed in partnership with Sourced, we're beginning to extract the full value of our data," says Loi.

Looking ahead, PayNet will continue to develop the platform's analytics capabilities with AWS, ensuring compliance with BNM directives on the use of cloud services in Malaysia's financial sector. Comments Tse, "By adhering to AWS security best practices with the support of Sourced, we implemented a zone-based approach and field-level encryption to minimize the risk of data exposure. This allows us to comply with BNM regulations while supporting the sustainable growth of digital banking services."

About PayNet

Payments Network Malaysia Sdn Bhd (PayNet) aims to build inclusive, accessible, and efficient payments and financial ecosystems in Malaysia. Based in Kuala Lumpur, the organization is helping the country accelerate its digitalization journey through world-class payment systems.

About Sourced Group

Sourced Group provides cloud computing services focused on security and compliance for regulated industries globally. It specializes in securing and managing cloud transformations, enhancing innovation and operational efficiency for enterprises.



Learn more

To learn more, visit aws.amazon.com/big-data/datalakes-and-analytics.

AWS Services Used

Amazon Managed Streaming for Apache Kafka

Amazon Managed Streaming for Apache Kafka (Amazon MSK) makes it easy to ingest and process streaming data in real time with fully managed Apache Kafka.

[Learn more »](#)

Amazon Elastic Kubernetes Service

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

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Amazon QuickSight

Amazon QuickSight powers data-driven organizations with unified business intelligence (BI) at hyperscale.

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